

LAB ACTIVITY 2

(Due September 23 @23:59)

Please submit the Jupiter notebook file with your answers. Please don't remove the error messages from the notebook.

1. Assuming there are no accidents or delays, the distance that a car travelled can be calculated using the formula:

Distance = speed * time

A car is travelling at 70 miles per hour. Write a program that displays the following: (3 points)

- a. The distance the car will travel in 6 hours
- b. The distance the car will travel in 10 hours

2. A car's miles-per-gallon can be calculated with the formula:

MPG = miles driven / gallons of gas used

Write a program that asks the user for the number of miles driven and the gallons of gas used. It should calculate the car's MPG and display the result. (3 points)

3. Write a program that asks the user to enter a number of seconds and then do the following: (5 points)

There are 60 seconds in a minute. If the number of seconds entered by the user is greater than or equal to 60, the program should convert the number of seconds to minutes and seconds.

4. A book club offers points to its customers based on the number of books purchased each month. The points are awarded as follows:

- if a customer purchases 0 books, he or she earns 0 points
- if a customer purchases 1 or 2 books, he or she earns 5 points
- if a customer purchases 3 or 4 books, he or she earns 15 points
- if a customer purchases 5 or 6 books, he or she earns 30 points
- if a customer purchases 7 or more books, he or she earns 60 points

Write a program that asks the user to enter the number of books he or she has purchased this month, and then displays the number of points awarded. (5 points)

5. Write a program that asks the user for a number in the range of 1 through 7. Your program should display the corresponding day of the week, where 1 = Monday, 2 = Tuesday, 3 = Wednesday, 4 = Thursday, 5 = Friday, 6 = Saturday and 7 = Sunday. The program should display an error message if the user enters a number that is outside the range. (5 points)